David Bedolla, PhD

Robotics Software Engineer | Real-Time Control | Motion | Hardware Deployment

Montréal, QC, Canada | davidbedollamartinez@outlook.es | linkedin.com/in/-davidbedolla
github.com/BedollaDavidPhD | orcid.org/0000-0001-8552-6842

PROFILE

Robotics software engineer with applied R&D experience in real-time control, robot dynamics, high-degree-of-freedom systems, and hardware deployment. Experience includes ROS 2 and C++ control systems, redundancy resolution, inverse kinematics, rehabilitation exoskeleton controllers, symbolic modeling, and sensor-based motion generation using IMU, EMG, and depth-camera data. Work connects software architecture, dynamic modeling, motion control, hardware integration, and physical validation for assistive robotics and mobile manipulation.

PROFESSIONAL EXPERIENCE

Associate Researcher

2026-Present | Lab INIT Robots, Montréal

- Deployed a 400 Hz whole-body control system for a 10-DoF mobile manipulator with joystick-based commands.
- Implemented Kinova Gen3 redundancy resolution with approximately 3 microseconds runtime on an Intel i7 processor.
- Integrated the method into a real-time ROS 2 and C++ pipeline using Pinocchio.
- Improved Cartesian tracking with a super-twisting controller and safety-oriented command logic.

Postdoctoral Researcher

2025 | ÉTS, Montréal

- Reduced lower-limb robot control latency by 30 times, enabling torque-based control and lower tracking error.
- Improved ROS and Python real-time scheduling for autonomous vehicle control, reducing steering oscillations.
- Developed symbolic dynamics and inverse-kinematics GUI tools for rapid robot modeling and validation.
- Supported international collaborations on wheelchair-mounted robots and upper-limb rehabilitation systems.

Research Professor

2024 | Universidad Tecnológica de la Mixteca, Mexico

- Developed and tested an optimization strategy for a 7-DoF robot, improving tracking performance by 20 percent.
- Supported the mechanical design of a three-finger adaptive robotic gripper for activities of daily living.
- Led a national student race-vehicle project through design, construction, testing, and competition.

Research Assistant

2020-2023 | ÉTS, Montréal

- Deployed learning-based inverse kinematics for a 7-DoF rehabilitation exoskeleton in approximately 43 microseconds.
- Developed a 1 to 4 kHz hard real-time motion-control stack and a robust predictive controller.
- Built LabVIEW and Simscape digital twins and integrated EMG, IMU, and depth-camera sensing.
- Designed an EMG and IMU mirror-rehabilitation system for upper-limb therapy.

Research Professor

2019 | Universidad de la Sierra Juárez, Mexico

- Implemented neural-network inverse kinematics for a 6-DoF PUMA robot.
- Developed a MATLAB convolutional neural network for wildlife detection in camera-trap datasets.

Professor

2017-2018 | Tecnológico Nacional de México, Mexico

- Deployed a real-time robotic manipulator emulator on a TI microcontroller with a 10 microsecond sampling period.
- Implemented an FPGA-based voice-processing system using linear predictive coding.

EDUCATION

2020-2023	**PhD in Robotics** Learning-Based Upper-Limb Robotic Rehabilitation, ÉTS, Montréal
2014-2016	**M.Sc. in Robotics** Hardware-in-the-Loop Simulation of a Robotic Manipulator, Universidad Tecnológica de la Mixteca
2009-2014	**B.Eng. in Mechatronics** Universidad Tecnológica de la Mixteca, CENEVAL EGEL Outstanding distinction

SELECTED PUBLICATIONS

- **Bedolla-Martínez, D.**, Kali, Y., Saad, M., Ochoa-Luna, C., and Rahman, M. H. (2023). *Learning human inverse kinematics solutions for redundant robotic upper-limb rehabilitation.* Engineering Applications of Artificial Intelligence, 126, 106966. DOI: 10.1016/j.engappai.2023.106966.
- **Bedolla-Martínez, D.**, Kali, Y., Saad, M., Ochoa-Luna, C., and Rahman, M. H. (2023). *Robust MPC with Integral Super-Twisting for Trajectory Tracking of an Exoskeleton Robot Arm.* IEEE PEDS. DOI: 10.1109/PEDS57185.2023.10246735.
- **Bedolla-Martínez, D.** (2023). *Learning-Based Upper Limb Robotic Rehabilitation.* Doctoral dissertation, ÉTS.

SELECTED PROJECTS

- **Whole-Body Mobile Manipulation:** 400 Hz ROS 2 and C++ coordination of a mobile base and Kinova Gen3 arm.
- **Dynamics Forge:** Interactive modeling and control platform for copters, drones, pendulums, cart-pole systems, and manipulators.
- **Automatic Robotics Tooling:** Symbolic kinematics, inverse kinematics, RNEA, and expression-optimization tools for serial and branched robots.
- **Learning-Based Rehabilitation:** Gaussian Process inverse kinematics, robust predictive control, and sensor-driven mirror therapy.

TECHNICAL SKILLS

Programming	C, C++, Python, MATLAB, LabVIEW, VHDL
Robotics	ROS 2, Pinocchio, URDF, motion planning, real-time control, mobile manipulation
Modeling and Control	Kinematics, dynamics, RNEA, MPC, sliding-mode control, reinforcement learning, Gaussian Processes
Simulation and Hardware	Simulink, Simscape Multibody, LabVIEW RT and FPGA, DAQ, microcontrollers, IMU, EMG, depth cameras